

Sodium–glucose cotransporter 2 inhibitors across cardio-reno-metabolic disease and body mass index: a meta-analysis of randomized controlled trials

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Aims

Obesity is a central driver of cardio-reno-metabolic (CRM) disease and may influence pharmacological responsiveness. Sodium–glucose cotransporter 2 inhibitors (SGLT2i) improve outcomes in heart failure (HF), chronic kidney disease (CKD), and type 2 diabetes (T2D), but whether body mass index (BMI) modifies these benefits across a broad CRM spectrum remains uncertain.

Methods and results

We performed a meta-analysis of randomized controlled trials (RCTs) comparing SGLT2i with placebo in patients with HF, CKD, and/or T2D and reported BMI-stratified outcomes using a cutoff of 30 kg/m². Outcomes included the composite of cardiovascular death or HF hospitalization and its components, and all-cause death. Random-effects meta-analyses were performed using risk ratios (RR). Eight RCTs encompassing 57 356 patients were included (mean BMI 29.95 kg/m²). SGLT2i significantly reduced the risk of cardiovascular death or HF hospitalization in patients with BMI <30 kg/m² (RR 0.80, 95% CI 0.72–0.89) and BMI ≥30 kg/m² (RR 0.80, 95% CI 0.75–0.85), with no evidence of effect modification by BMI, an effect primarily driven by reductions in heart failure hospitalizations. Consistent benefits were observed for HF hospitalization (RR 0.75 for BMI <30 kg/m² and 0.74 for BMI ≥30 kg/m²) and all-cause mortality (RR 0.83 for BMI <30 kg/m² and 0.89 for BMI ≥30 kg/m²). Meta-regression identified disease category and SGLT2i molecule as sources of heterogeneity, without altering the absence of BMI interaction. Sensitivity, cumulative, and bootstrap analyses confirmed the robustness of the results.

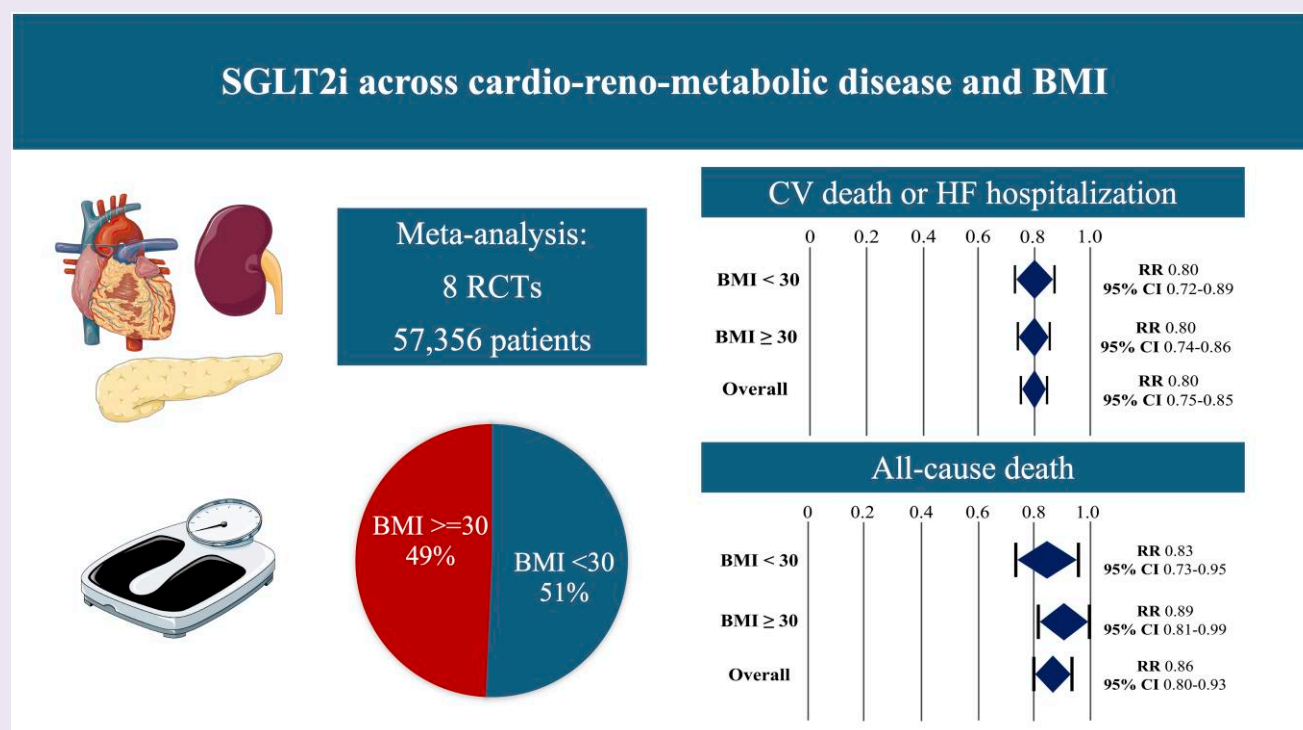
Conclusion

Across a broad CRM population, SGLT2i reduce cardiovascular death or HF hospitalization and all-cause death irrespective of BMI.

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Graphical abstract



In the present meta-analysis of eight randomized controlled trials encompassing 57 356 patients across the cardio-reno-metabolic spectrum, including heart failure (HF), chronic kidney disease (CKD), and/or type-2 diabetes, 49% of patients had obesity, as defined by a body mass index (BMI) ≥ 30 kg/m. Sodium-glucose cotransporter-2 inhibitors reduced significantly the risk of the composite of cardiovascular (CV) death or HF hospitalization and all-cause death compared to placebo consistently in patients with or without obesity.

Keywords

Sodium-glucose cotransporter 2 inhibitors • Body mass index • Obesity • Heart failure • Chronic kidney disease • Diabetes

Introduction

Cardio-reno-metabolic (CRM) disease refers to the convergence of cardiovascular, renal, and metabolic conditions that share common risk factors and pathophysiological mechanisms and that mutually amplify each other's incidence, progression, and adverse outcomes.¹ The clustering of heart failure (HF), chronic kidney disease (CKD), and type-2 diabetes (T2D) accounts for nearly one-third of mortality and inpatient costs.^{2,3} This burden stresses the need for therapeutic strategies able to address the risk across the whole CRM spectrum.

Sodium-glucose cotransporter 2 inhibitors (SGLT2i) emerged as the first class of drugs exerting consistent disease-modifying effects across the CRM spectrum. Initially developed as antidiabetic agents, SGLT2i have demonstrated robust benefits in reducing HF hospitalizations, slowing CKD progression, and decreasing major adverse cardiovascular events in patients with T2D, HF, and CKD.⁴ These effects appear to extend far beyond their glucose-lowering effect, potentially involving multiple haemodynamic, renal, and metabolic mechanisms.⁵

Obesity is a central driver of CRM conditions. Excess adiposity promotes HF, CKD, and T2D through a combination of metabolic, haemodynamic, and inflammatory mechanisms.⁶ The

prevalence of obesity is continuously rising in Europe,⁷ while obesity- and HF-related mortality has tripled over the past 25 years in the US.⁸ Obesity can alter pharmacokinetics and pharmacodynamics and therefore modify responsiveness to drug therapies, raising the question of whether drugs have consistent effects across body mass index (BMI) categories.⁹

Subgroup analyses of individual randomized controlled trials (RCTs) in patients with HF, CKD, or T2D have shown that the cardioprotective and/or renoprotective effects of SGLT2i are consistent across BMI categories in each one of these conditions. A previous meta-analysis has further confirmed these findings across HF studies.¹⁰ However, no comprehensive meta-analysis has evaluated the impact of BMI on the effects of SGLT2i across the broad CRM spectrum. To address this gap in knowledge, we conducted a meta-analysis of RCTs assessing the effects of SGLT2i in diverse CRM populations by BMI category.

Methods

This meta-analysis was conducted and reported in adherence to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.¹¹ No ethics committee approval

was required as this study involved a retrospective analysis of publicly available data.

Criteria for eligibility and study selection

All included studies were RCTs including randomized patients with HF, CKD, and/or T2D into SGLT2i vs. placebo. Trials were categorized according to their primary enrolled population, while no patient-level data were available to define multimorbidity patterns or formal CRM staging within individual participants. Each study was required to have a BMI-stratified analysis of the primary outcomes, including at least one of the following: all-cause mortality, cardiovascular mortality, and/or HF hospitalization. The studies were also required to stratify patients according to BMI into at least two categories, with the preselected cutoff value of 30 kg/m². The predefined combined effect size was the risk ratio, and therefore, studies that did not provide raw event counts, rendering risk ratios extractable, were excluded. Observational non-randomized studies were also excluded. After the initial study selection, the snowball sampling approach was employed to identify additional relevant articles, which did not yield any additional studies.

Search strategy

An electronic search was performed on 20 December 2025, by two researchers (DF and KK) using single terms and phrases through four databases, such as PubMed (<https://www.pubmed.ncbi.nlm.nih.gov/>), Scopus (<https://www.scopus.com/>), Web of Science (<https://www.webofscience.com/>), and Cochrane (<https://www.cochranelibrary.com/>) for articles published in the English language. For PubMed, the predefined searching strategy was the following: (((((((canagliflozin) OR (dapagliflozin)) OR (empagliflozin)) OR (ertugliflozin)) OR (sodium–glucose cotransporter 2 inhibitor)) OR (SGLT2 inhibitor)) AND (((((((cardiorenometabolic) OR (heart failure)) OR (chronic kidney disease)) OR (type 2 diabetes)) OR (diabetes)) OR (renal)) OR (renal dysfunction))) AND (((((((cardiovascular outcome) OR (major adverse cardiovascular event)) OR (mortality)) OR (MACE)) OR (cardiovascular death)) OR (myocardial infarction)) OR (stroke)) OR (heart failure hospitalization)) OR (all-cause mortality))) AND (('2015/01/01'[Date–Publication] : '3000'[Date–Publication])). The complete search strategy is analysed in [Supplementary material online, Table S1](#), for the rest of the databases.

Data extraction

The results from the predefined searching strategy of different databases were collectively imported to EndNote™ software, version 21, and each level of screening was performed by two independent researchers (DF and KK), based on the previously analysed inclusion and exclusion criteria. After removal of duplicates, screening was conducted based on title and abstract reviewing, followed by full-text screening of the remaining studies. Additional details about the screening process are provided in the flowchart ([Figure 1](#)).

Using a preconfigured template, data were independently extracted by two investigators from the included studies. Extracted data included: study characteristics (author, study design, year of publication, disease category, drug molecule category, and drug dosage), population characteristics (patient number, age, sex, mean follow-up, T2D percentage, female percentage, mean eGFR, mean BMI, and mean LVEF), and primary outcomes (cardiovascular mortality and/or HF hospitalization, all-cause mortality per treatment arm, and BMI category), as seen in [Table 1](#). When trials reported multiple BMI categories, adjacent categories were merged by summing participants and events to construct the prespecified dichotomy (<30 vs. ≥30 kg/m²). If published BMI stratification could not be mapped to this cutoff, the trial was not included in quantitative synthesis for that outcome. The results were cross-verified and reached through a consensus discussion between two researchers. If data were not available for some articles, we contacted the corresponding author up to three times before excluding them if we received no response. For studies with multiple publications, we compared the reports and selected the one with the most recent and comprehensive data.

Quality assessment

We used the Cochrane Collaboration's tools RoB 2.0¹² to assess the risk of bias among included RCTs ([Supplementary material online, Figure S1](#)). Two researchers (DF and KK) independently performed the assessments, while any discrepancies were resolved through consultation with the study supervisor (GF).

Statistical analysis

All data were analysed using the R statistical software, version 4.2.0,¹³ with the 'meta', 'robmeta', and 'metafor' packages.¹⁴ The combined effect sizes for each outcome were expressed as risk ratios (RR) with corresponding standard errors (SE) and 95% confidence intervals (CI). All were calculated using a random-effects model with the restricted maximum-likelihood (REML) adjustment. We selected the random-effects model due to the presumable clinical heterogeneity in our data, which arose from variations in patient characteristics, as well as within study characteristics, sample size, follow-up, drug molecule category, and disease category.

Heterogeneity was assessed using the chi-square (Q) statistic, which reflects total dispersion, and the I² statistic, following Higgins' method, which indicates the percentage of excess dispersion relative to total dispersion. The tau-squared (T²) parameter was also estimated as an absolute measure of heterogeneity. In cases of high heterogeneity (namely I² > 50%), it was intended to perform meta-regression analysis to identify potential variables that might explain the observed heterogeneity. The following study- and population-specific variables were considered: year of publication, follow-up duration, drug molecule, disease category, total population size, placebo event rate (used as a surrogate for baseline population risk), percentage of T2D, female percentage, mean eGFR, mean BMI, mean LVEF, and mean age. Variables that significantly explained the within study heterogeneity were used to perform subgroup analyses and assess possible effect modifications. Notably, given the limited number of the finally included trials, meta-regression analyses were considered exploratory and hypothesis-generating. Additionally, publication bias was visually tested by inspection of the Funnel plots, while it was also statistically tested by performing Egger's and Begg-Mazumdar tests. In case of statistically significant publication bias, the trim-and-fill method was intended to be applied to correct for Funnel plot asymmetry.

To assess the robustness and internal validity of the meta-analysis results and the influence of individual studies on the overall results, leave-one-out sensitivity analyses—where each study was sequentially removed—and cumulative meta-analyses—where studies were sequentially added from the earliest to the most recent—were performed. For the same purpose, bootstrap resampling (10 000 iterations) was performed to estimate the combined risk ratio, Q-statistic, and I² for each primary outcome.

Last, to formally assess the possible effect modification by BMI, interaction was evaluated using the test for subgroup differences within the random-effects framework. Specifically, log risk ratios were compared between BMI <30 kg/m² and BMI ≥30 kg/m² strata, and a P-value for interaction was calculated for each outcome. Additionally, the difference between log risk ratios in the BMI ≥30 kg/m² and BMI <30 kg/m² strata within each trial was calculated ($\Delta\log RR = \log RR_{\geq 30} - \log RR_{< 30}$), and the corresponding standard error was derived assuming independence between strata. These within-trial interaction estimates were pooled using a random-effects model (REML), and the results were expressed as the ratio of risk ratios ($RRR = \exp(\Delta\log RR)$) with 95% confidence intervals.

All statistical analyses were conducted independently for each outcome and BMI category.

Effect size selection

RRs were selected as the primary combined effect measure for quantitative synthesis, based on methodological and practical considerations. First, included studies exhibited variable follow-up durations, limiting the comparability of hazard ratios (HRs), which are inherently dependent on time-to-event analyses and underlying survival distributions. Second, HR estimates assume proportional

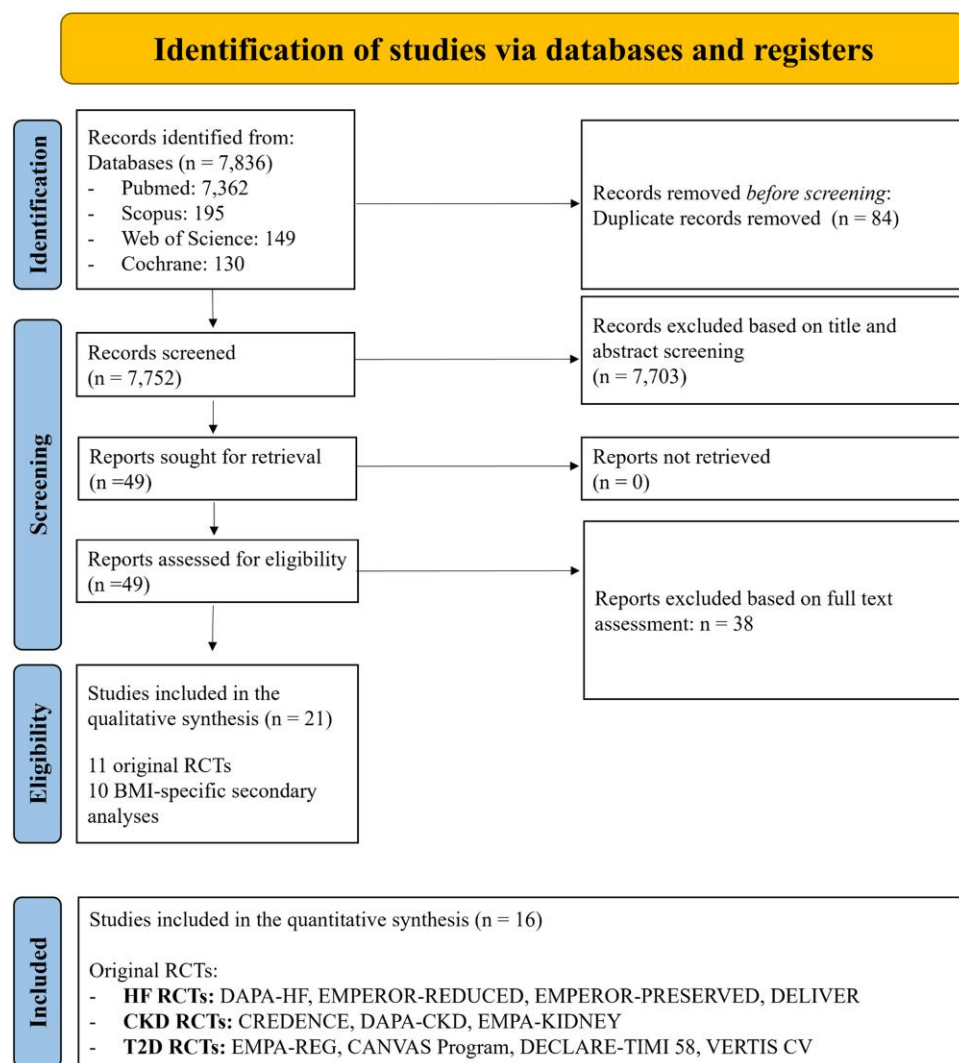


Figure 1 Preferred reporting items for systematic reviews and meta-analyses flowchart.

hazards over the entire follow-up period, an assumption that may not be consistently satisfied across heterogeneous study populations. Third, HRs were not uniformly reported across all eligible studies, as opposed to RRs, enabling a more comprehensive and methodologically consistent meta-analysis. Additionally, RRs were preferred because HRs were typically reported across more granular BMI categories and were not mergeable for the <30 vs. ≥ 30 kg/m² strata. Last, sensitivity analyses were performed for studies with available HRs for cardiovascular death and HF hospitalization, and BMI subgroups were available to evaluate the robustness of results derived from RR-based analyses.

Results

Search results

A total of 7836 studies were initially retrieved from all databases (PubMed, Scopus, Web of Science, and Cochrane). After removing duplicates, a total of 7752 articles remained. After title and abstract screening, a total of 49 articles were considered eligible for full-text assessment, while 21 of them were eventually

selected after full-text screening. Eleven RCTs were included in the qualitative synthesis, while three were excluded from the quantitative meta-analysis due to the absence of BMI-specific secondary analyses necessary for BMI categorization and outcome assessment. Finally, eight RCTs and their BMI-specific secondary analyses were included in this meta-analysis (Figure 1). Among the 11 included studies, 4 included primarily patients with HF,¹⁵⁻¹⁸ 3 patients with CKD,¹⁹⁻²¹ and 4 patients with T2D.²²⁻²⁵

Study characteristics

The eight RCTs included in the quantitative meta-analysis encompassed a total of 57 356 patients with CRM conditions who were randomized into SGLT2i vs. placebo and were followed up for a median of 28.2 months (Table 1). Among them, 28 248 (49.2%) had a BMI <30 , and 29 108 (50.8%) a BMI ≥ 30 . Patients had a median age of 65.35 years, a mean reported LVEF of 42.65%, a mean eGFR of 63.9 mL/min/1.73 m², and a mean BMI of 29.95, suggesting balanced distributions based on the cutoff value of 30.

Table 1 Continued

Study	Year	Condition	Drug	Population ^a	Age	% female	BMI (mean ± SD)	% with T2D	eGFR (mean)	Follow-up (median, months)	Primary endpoint	Urgent HF visits included	BMI-stratified primary outcome data available
VERTIS CV	2020	T2D	Ertugliflozin 5 or 15 mg	8243	64.4	30	31.93 ± 5.43	100	76	42	CV death, nonfatal myocardial infarction or nonfatal stroke	No	Yes (quantitative synthesis)

BMI, body mass index; T2D, type-2 diabetes; eGFR, estimated glomerular filtration rate in mL/min/1.73 m²; HF, heart failure; CV, cardiovascular; CKD, chronic kidney disease.
^aRefers to the population included in BMI-specific secondary analyses of randomized controlled trials.

All eight studies reported events for the combined endpoint of cardiovascular mortality and HF hospitalization, while six of them reported all-cause mortality, five reported cardiovascular mortality, and five reported HF hospitalizations (Table 1). Based on the BMI-stratified secondary analyses, a total of 6039 events of the combined endpoint of cardiovascular mortality and/or HF hospitalization were reported, while a median of 187 events occurred for patients with a BMI <30 and 174 events for those with a BMI ≥30 (Table 1).

Quality assessment

All studies were assessed to have a low risk of bias, with excellent performance in all risk of bias domains, including the randomization process, deviations from the intended intervention, missing outcome data, outcome measurement, and selection of the reported result (Supplementary material online, Figure S1).

Main results

Cardiovascular death and HF hospitalization

The combined risk ratio for cardiovascular death or HF hospitalization by SGLT2i vs. placebo was 0.80 (95% CI, 0.72–0.89) for patients with BMI <30 kg/m² and 0.80 (95% CI, 0.74–0.86) for those with BMI ≥30 kg/m², with no significant effect modification between BMI subgroups (Figure 2). Regarding publication bias, the Funnel plots inspection revealed a relative symmetry (Supplementary material online, Figure S2A and S2B), which were confirmed by the Egger’s test for both BMI categories (P = 0.87 and 0.74). For that reason, no trim-and-fill analysis was performed. However, given the limited number of studies, the results of publication bias are not always reliable and should be interpreted cautiously. Formal testing for interaction using the test for subgroup differences did not demonstrate statistically significant effect modification by BMI (P for interaction = 0.98). Additionally, the pooled ratio of risk ratios (BMI ≥30 vs. <30) was RRR = 0.99 (95% CI 0.86–1.15; P = 0.91), also confirming non-significance.

Significant heterogeneity was observed for the subgroup of patients with BMI <30 kg/m² (I² = 54.2%, Q-statistic = 15.29, P = 0.03). Exploratory meta-regression analyses suggested that disease category and SGLT2 inhibitor molecule were significantly associated with between-study heterogeneity in the composite outcome of cardiovascular death and HF hospitalization across BMI categories (Supplementary material online, Table S2). Specifically, in patients with BMI <30 kg/m², dapagliflozin showed a statistically significant association with improved outcomes (RR 0.88, 95% CI, 0.79–0.97, P = 0.009), whereas other SGLT2i did not. Disease category further contributed to heterogeneity in BMI <30 kg/m², with significant associations observed in CKD and HF populations. Similarly, among patients with BMI ≥30 kg/m², disease category significantly explained heterogeneity, particularly in subgroups with CKD and HF, where SGLT2i were associated with a reduced risk of the composite endpoint. Drug-specific effects were also evident in this BMI group (Supplementary material online, Table S2). No significant effect modification was observed for all other variables in either BMI category. However, these findings should be interpreted cautiously, given the limited number of trials included in the meta-regression models, which reduces statistical power and limits interpretability.

The composite endpoint was significantly reduced across BMI strata; this effect was primarily driven by reductions in HF hospitalization. Notably, the reduction in cardiovascular death alone was directionally consistent but did not reach statistical

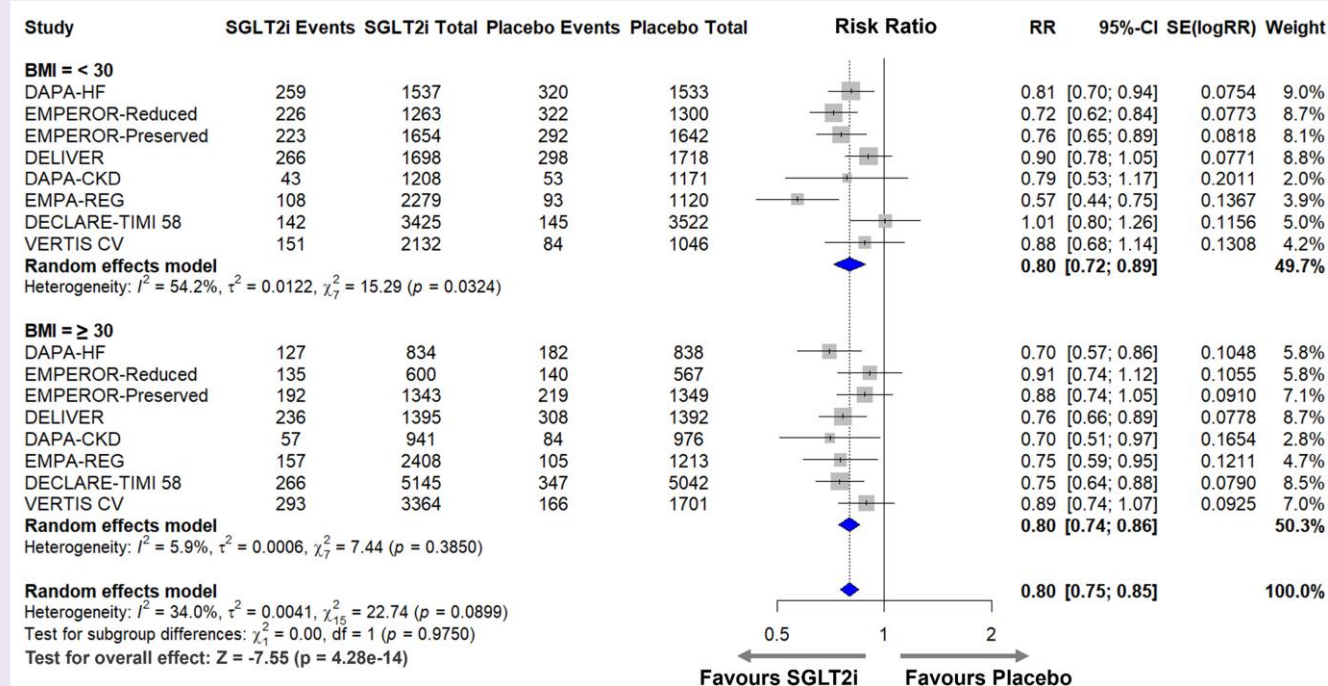


Figure 2 Forest plot of SGLT2 inhibitor efficacy on the composite endpoint of cardiovascular death or heart failure hospitalization in patients with cardio-reno-metabolic disease according to BMI categories.

significance within individual BMI categories. Importantly, although relative treatment effects were consistent across BMI strata, absolute benefits varied according to baseline disease risk, with higher-risk populations such as heart failure showing greater numbers of events prevented per 1000 treated patients (Supplementary material online, Table S4).

All-cause death

In six studies reporting BMI-stratified data on SGLT2i efficacy on all-cause death, the combined risk ratio was 0.83 (95% CI, 0.73–0.95) for patients with BMI <30 kg/m² and 0.89 (95% CI, 0.81–0.99) for those with BMI ≥30 kg/m², with no significant effect modification between BMI subgroups (Figure 3). Regarding publication bias, the Funnel plot inspection revealed a relative asymmetry for BMI <30 (Supplementary material online, Figure S2C), which was confirmed by the Egger's test ($P = 0.02$). The number of imputed studies indicated by the trim-and-fill analysis was zero, indicating an inconclusive result. For BMI ≥30, the Funnel plot inspection revealed symmetry (Supplementary material online, Figure S2D) and was confirmed by the non-significant Egger's test ($P = 0.48$). Notably, the limited number of studies limits the interpretability of the publication bias results. Formal testing for interaction using the test for subgroup differences did not demonstrate a statistically significant effect modification by BMI (P for interaction = 0.39). Additionally, the pooled ratio of risk ratios (BMI ≥30 vs. <30) was RRR = 1.04 (95% CI 0.92–1.18; $P = 0.49$), also confirming non-significance.

Significant heterogeneity was observed for the subgroup of patients with BMI <30 kg/m² ($I^2 = 59.2\%$, Q -statistic = 12.26, $P = 0.03$). Meta-regression analyses demonstrated that both drug molecule and disease category significantly explained between-study heterogeneity for all-cause death across BMI categories (Supplementary material online, Table S3). In patients with BMI

<30 kg/m², empagliflozin showed a significant association with improved outcomes (RR 0.77, 95% CI, 0.60–0.99, $P = 0.009$), whereas other SGLT2i did not. Disease category further contributed to heterogeneity in this BMI group, with significant associations observed in CKD populations. No significant effect modification was observed for all other variables in either BMI category.

Cardiovascular death

In five studies reporting BMI-stratified raw counts of cardiovascular death, the combined risk ratio was 0.91 (95% CI, 0.82–1.02) for patients with BMI <30 kg/m² and 0.90 (95% CI, 0.80–1.01) for those with BMI ≥30 kg/m², with no significant effect modification between BMI subgroups (Figure 4). Regarding publication bias, the Funnel plots inspection revealed a relative symmetry (Supplementary material online, Figure S2E and S2F), confirmed by the Egger's test for both BMI categories ($P = 0.22$ and 0.74). Notably, the limited number of studies limits the interpretability of the publication bias results. No significant heterogeneity was calculated in either BMI category. Formal testing for interaction using the test for subgroup differences did not demonstrate a statistically significant effect modification by BMI (P for interaction = 0.84). Additionally, the pooled ratio of risk ratios (BMI ≥30 vs. <30) was RRR = 0.96 (95% CI 0.81–1.13; $P = 0.61$) also confirming non-significance.

HF hospitalization

In five studies reporting BMI-stratified raw counts of HF hospitalization, the combined risk ratio was very significant at 0.75 (95% CI, 0.64–0.87) for patients with BMI <30 kg/m² and 0.74 (95% CI, 0.68–0.82) for those with BMI ≥30 kg/m², with no significant effect modification between BMI subgroups (Figure 5). Regarding publication bias, the Funnel plots inspection revealed

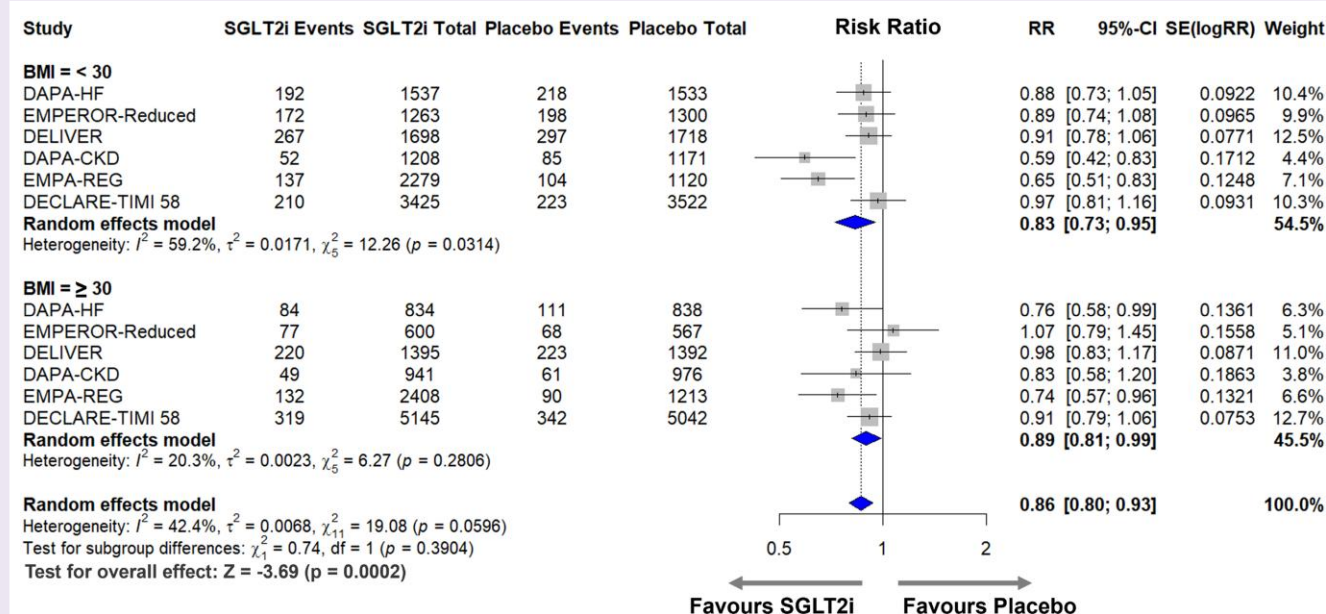


Figure 3 Forest plot of SGLT2 inhibitor efficacy on all-cause death in patients with cardio-reno-metabolic disease according to BMI categories.

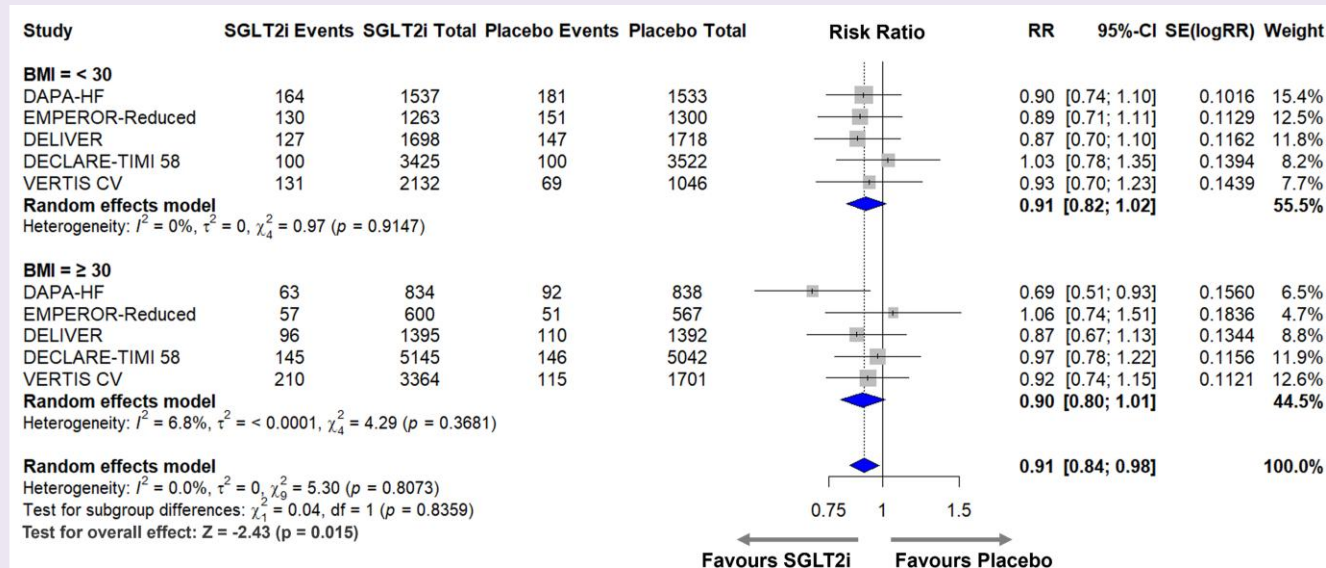


Figure 4 Forest plot of SGLT2 inhibitor efficacy on cardiovascular death in patients with cardio-reno-metabolic disease according to BMI categories.

a relative symmetry (Supplementary material online, Figure S2G and S2H), confirmed by the Egger's test for both BMI categories ($P = 0.67$ and 0.61). Notably, the limited number of studies limits the interpretability of the publication bias results. No significant heterogeneity was calculated in either BMI category. Formal testing for interaction using the test for subgroup differences did not demonstrate a statistically significant effect modification by BMI (P for interaction = 0.94). Additionally, the pooled ratio of risk ratios (BMI ≥ 30 vs. < 30) was $RRR = 0.99$ (95% CI 0.80 – 1.22 ; $P = 0.92$), also confirming non-significance.

Sensitivity and cumulative analyses

The pooled effect estimate for all outcomes did not change direction in leave-one-out sensitivity analyses (Supplementary material online, Figure S3A, S3C, S4A, S4C, S5A, S5C, S6A, S6C), indicating the absence of influential single-study effects across all outcomes and BMI subgroups. Likewise, the temporal cumulative analyses for all outcomes revealed that the sequential addition of studies from the earliest to the most recent demonstrated consistent and stable combined effect size direction (Supplementary

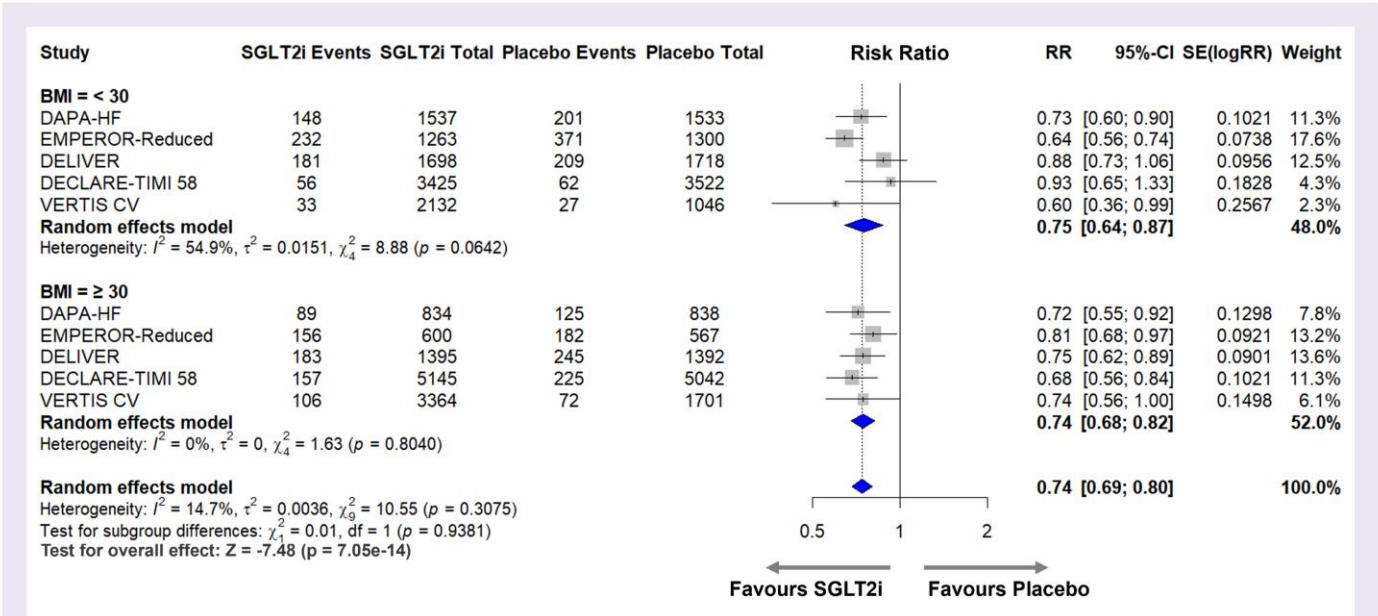


Figure 5 Forest plot of SGLT2 inhibitor efficacy on heart failure hospitalization in patients with cardio-reno-metabolic disease according to BMI categories.

material online, Figure S3B, S3D, S4B, S4D, S5B, S5D, S6B, S6D). These results collectively demonstrate robustness of the meta-analysis findings across all outcomes and BMI categories, with no significant single-study or temporal effects.

Sensitivity analyses using available HRs—only available for HF RCTs—also yielded directionally and statistically consistent results with the primary RR-based analyses, supporting the robustness of the main findings (Supplementary material online, Figure S11A). Similarly, a sensitivity analysis restricted to trials with closely matching endpoint definitions, categorizing them based on the inclusion of urgent HF visits, demonstrated results consistent with the primary analysis, with unchanged direction and magnitude of treatment effect and no significant interaction by BMI (Supplementary material online, Figure S11B).

Bootstrap analysis

For the composite of cardiovascular death or HF hospitalization, it was shown that the 95% CI of the pooled RR distribution were ranging from 0.77 to 0.83 for BMI <30 kg/m² (median 0.80) and from 0.77 to 0.83 for BMI ≥30 kg/m² (median 0.80), confirming the similarly significant results across BMI categories (Supplementary material online, Figure S7A and S7D). Heterogeneity metrics were consistent with prior conclusions based on the initial meta-analyses (Supplementary material online, Figure S7B, S7C, S7E, S7F).

For all-cause mortality, it was shown that the 95% CI of the pooled RR distribution were ranging from 0.80 to 0.91 for BMI <30 kg/m² (median 0.86) and from 0.80 to 0.92 for BMI ≥30 kg/m² (median 0.86), confirming the similarly significant results across BMI categories (Supplementary material online, Figure S8A and S8D). Heterogeneity metrics were consistent with prior conclusions based on the initial meta-analyses (Supplementary material online, Figure S8B, S8C, S8E, S8F).

Impact of BMI across subgroups

According to the results of the meta-regression analyses of cardiovascular death or heart failure hospitalization and all-cause

death, subgroup analyses were performed based on disease category (HF, CKD, and T2D) and drug molecule (dapagliflozin, empagliflozin, and ertugliflozin). For both outcomes, subgroup analyses by disease category demonstrated largely consistent treatment effects across different disease population groups (Supplementary material online, Figures S9A, S9B, S10A, S10B).

On the other hand, for the composite of cardiovascular mortality or heart failure hospitalization, in patients with BMI <30 kg/m², the combined effect size was 0.71 (95% CI, 0.63–0.79) for empagliflozin and 0.87 (95% CI, 0.79–0.97) for dapagliflozin (Supplementary material online, Figure S9C). Although established statistical methods for comparing subgroup pooled effect sizes using statistical inference have important limitations, the 95% CI of the two subgroup-combined effect sizes present no overlap, with a statistically significant test for subgroup differences ($P = 0.01$). Conversely, in patients with BMI ≥30 kg/m², the combined effect size was 0.86 (95% CI, 0.76–0.96) for empagliflozin and 0.74 (95% CI, 0.68–0.81) for dapagliflozin (Supplementary material online, Figure S9D). Given the limitations of subgroup analyses, these findings suggest only a hypothesis-generating potential for BMI-dependent effect modification according to the specific SGLT2i molecule, with empagliflozin demonstrating greater protection in BMI <30 kg/m² and dapagliflozin conferring greater protection in BMI ≥30 kg/m². These trends across different SGLT2i molecules were not observed when assessing all-cause mortality (Supplementary material online, Figure S10C and S10D).

Discussion

The present meta-analysis, pooling randomized controlled data on more than 57 000 patients across the spectrum of CRM disease, shows that SGLT2i reduces the risk of cardiovascular events and all-cause death irrespective of baseline BMI. Notably, the observed benefit in the composite endpoint was primarily attributable to reductions in HF hospitalization. Specifically, the totality of evidence in patients with HF, CKD, and/or T2D demonstrates that SGLT2i reduce the risk of

cardiovascular death or HF hospitalization by 20%—an effect mainly driven by a 25% reduction in the risk of HF hospitalization—as well as the risk of all-cause death by 14%. Importantly, these benefits are consistent in patients with BMI <30 kg/m² and in those with BMI ≥30 kg/m², although absolute risk reductions were greater in higher-risk populations such as heart failure. These findings confirm that the disease-modifying benefits of SGLT2i across CRM disease are not modified by the presence of obesity, as defined by BMI.

The present work approach extends the findings from subgroup analyses of individual trials or from meta-analyses on specific CRM components, such as HF,¹⁰ by integrating HF, CKD, and T2D into a single framework. Conceptually, this approach provides a holistic perspective reflecting real-world multimorbidity, as patients often manifest concurrently or develop these interrelated conditions sequentially. In fact, one-fourth of patients with T2D have HF as their first complication,²⁶ while nearly 60% of patients with T2D will develop CKD.²⁷ From a clinical viewpoint, these findings are important, given that obesity is highly prevalent among patients with CRM disease and is often associated with therapeutic uncertainty due to altered drug pharmacokinetics and comorbidity burden or bias towards the selection of specific medications.^{6,28} Notably, the term CRM spectrum is used conceptually to denote the shared pathophysiological continuum encompassing HF, CKD, and T2D, as defined by the populations enrolled in the included randomized controlled trials. Patient-level multimorbidity, formal CRM staging, or progressive CRM burden were not assessed but rather trial-level populations representing major components of the CRM continuum were examined.

In all studies included in this meta-analysis, obesity was defined as a BMI ≥30 kg/m². This is a simple and practical way to define obesity, but it does not take into account the body composition and the relative contribution of fat and lean mass to the body weight.²⁹ In fact, the definition of obesity affects its epidemiology and prognostic impact. We observed a 49% prevalence of obesity, as defined by BMI. Similarly, in a pooled analysis of the five most prominent HFpEF drug trials, using only the BMI definition yielded an obesity prevalence of 47%.³⁰ However, adding to the definition also waist circumference and waist-to-height ratio, increased the prevalence of obesity to 95%.³⁰ In the same study, although BMI had a U-curve relationship with mortality, supporting the so-called 'obesity paradox,' this was not the case when BMI was substituted by the waist-to-height ratio, which had a continuous positive correlation with mortality.³⁰ These findings suggest that indices of central adiposity may better reflect cardiometabolic risk than BMI alone by providing improved biological resolution with regard to treatment responsiveness.³¹ Besides the total fat mass, the distribution of adipose tissue is also relevant. Different adipose tissue locations, including visceral, epicardial, ectopic and subcutaneous have differential metabolic impact.³² Therefore, dichotomization using a BMI threshold may obscure potential dose-response relationships between adiposity and therapeutic benefit and limit mechanistic inference regarding obesity as a potential modifier of treatment response. Future investigations should evaluate SGLT2i effects using more granular adiposity phenotypes, including anthropometric indices of central obesity and imaging-derived measures of adipose tissue distribution, helping refining risk stratification and improving the mechanistic understanding of treatment responsiveness across the CRM continuum.²⁹ Beyond anthropometric measures, metabolic health has emerged as a key determinant of cardiovascular risk, with adverse outcomes driven by metabolic

dysfunction irrespective of BMI category. In this context, the consistent benefit of SGLT2i across BMI strata supports the concept that therapeutic responsiveness may be mediated by underlying metabolic pathways rather than body size alone.³³

In this meta-analysis, we assessed the effects of SGLT2i in patients with CRM disease as defined by HF, CKD, and/or T2D, based on available RCTs. We were not able to assess the effects of SGLT2i on additional conditions of the CRM spectrum, such as ASCVD or atrial fibrillation. However, these conditions were highly prevalent in the patient populations of the trials included in this meta-analysis. For example, all patients enrolled in the EMPA-REG OUTCOME trial of empagliflozin in T2D had established ASCVD.²² Additionally, a recent meta-analysis showed that SGLT2i reduced the risk of major cardiovascular events, including cardiovascular death, myocardial infarction, or stroke, in patients with T2D and either established ASCVD or at high risk for ASCVD.⁴ Similarly, atrial fibrillation, closely interrelated to HF, CKD, obesity, and T2D,³⁴ was prevalent in 52% of patients with HFpEF and in 33% of patients with HFrEF randomized to empagliflozin or placebo in the EMPEROR-Preserved and EMPEROR-Reduced trials, respectively.^{35,36} Importantly, the beneficial effects of SGLT2i in HF were consistent in patients with and without atrial fibrillation.^{35,36} Non-alcoholic fatty liver disease (NAFLD) or metabolic dysfunction-associated steatotic liver disease (MASLD) is the hepatic component of CRM disease, being associated with obesity and type 2 diabetes and conferring an increased risk of incident HF and CKD.^{37,38} Although NAFLD-specific outcomes were not reported in the trials included in this meta-analysis, emerging evidence from clinical trials and meta-analyses shows that SGLT2i improve hepatic enzymes, imaging scores, and fibrosis indices in patients with T2D.^{39,40}

Moderate heterogeneity was observed for the composite outcome of cardiovascular death or HF hospitalization among patients with BMI <30 kg/m². Exploratory meta-regression analyses identified disease category and SGLT2i molecule as significant contributors to between-study heterogeneity. However, given the limited number of trials included in the meta-regression analyses and the reduced statistical power, these associations should be considered exploratory rather than definitive. Regarding disease category, subgroup analyses demonstrated consistent benefits across HF, CKD, and T2D populations. Regarding SGLT2i molecules, exploratory analyses suggested potential differences between individual SGLT2i according to BMI category; however, these findings do not suggest effect modification and should be interpreted cautiously. Randomized head-to-head comparisons between SGLT2i are lacking, and observed differences may reflect diverse trial design, patient selection, background therapies, or chance rather than true pharmacological divergence.

The present meta-analysis included large, high-quality RCTs with a low risk of bias and applied rigorous statistical methods that support the robustness and internal validity of the findings. However, it also has limitations. It is a study-level and not a patient-level meta-analysis, lacking therefore significant granularity. RCTs not providing outcomes by BMI category were inevitably excluded. In addition, the reported findings of subgroup analyses by disease and drug categories should be interpreted with caution, given their inherent limitations, including the risk of selection bias, reduced statistical power, and multiple comparisons.

In conclusion, SGLT2i improve cardiovascular outcomes and all-cause mortality across a broad spectrum of CRM disease irrespective of the presence of obesity as defined by BMI.

Further exploration of the effects of these drugs on additional CRM components and evaluation of the consistency of their effects using more granular definitions of obesity are warranted.

Supplementary material

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Data availability

All data are incorporated into the article and its online supplementary material.

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